

AKADEMIKA

VT- DTH

DTH TRAINER KIT

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. **AUDIO VIDEO ENGINEERING** is one of the important subject need to teach while learning electronics.

Whether satellite system is for transmitting voice, video & audio or training regardless of application, it is our vision to provide you with the product you need, ensuring lasting reliability & quality

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the Connecting cables are connected correctly as per experiment.

This system must be employed only for the use, for which it has been conceived, i.e. as educational system and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer system, turn off the power supply and do not tamper the system. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This system is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations.

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and upgradation of products, be sure to contact our service center or sales outlet.

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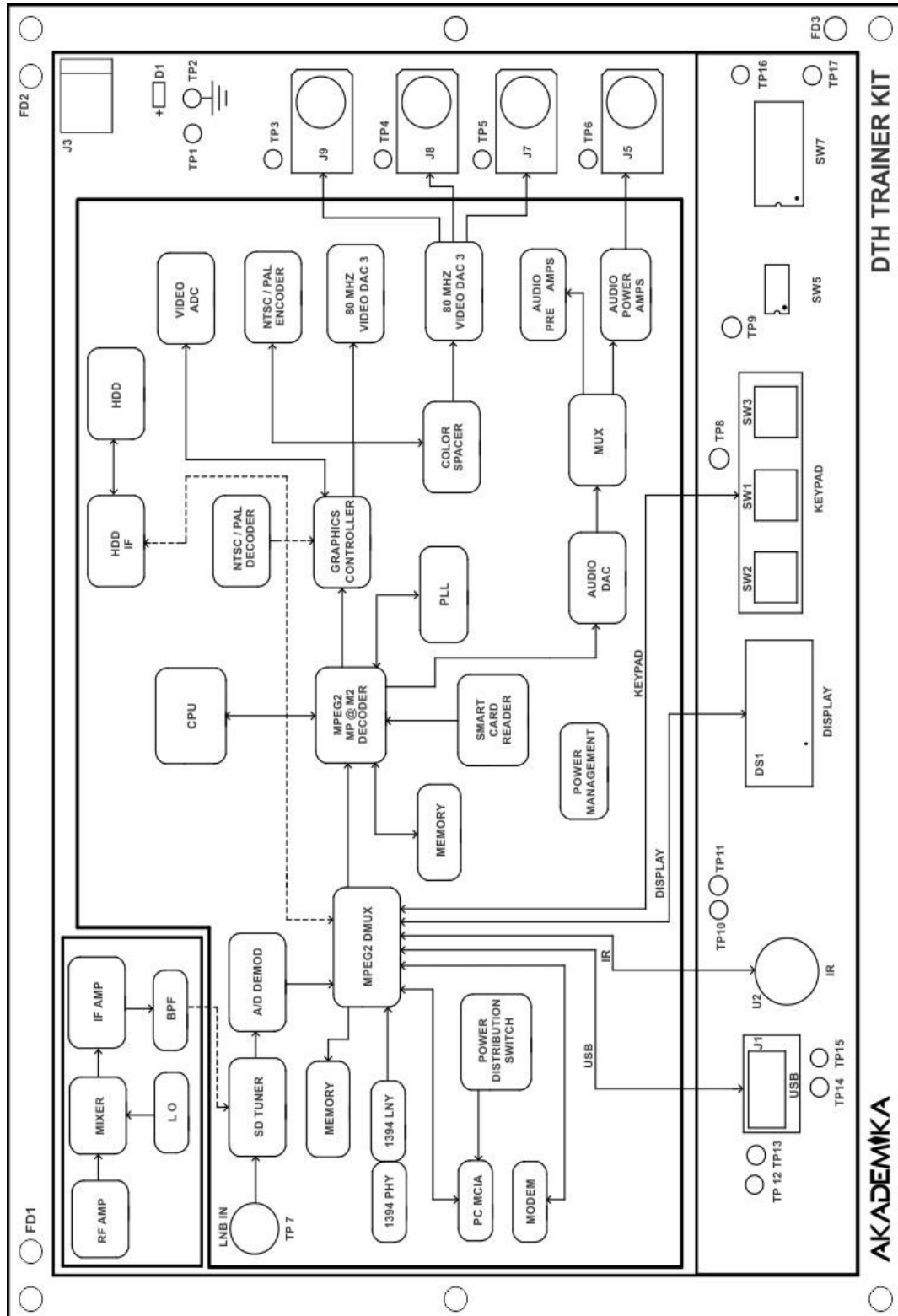
TECHNICAL SPECIFICATIONS

RF input	:	950-2150 MHZ
Input impedance	:	75Ω
Dish Input	:	F- Connector
Composite video output	:	RCA type
Audio output	:	RCA type (L/R)
IF Frequency	:	70 MHZ
Controls	:	UP/DOWN, Power ON/OFF, Switch Faults etc.
Test Points	:	17 NO
Switch Fault	:	11 NO
Power supply	:	12V Adaptor
Video Quality	:	1080P Full HD

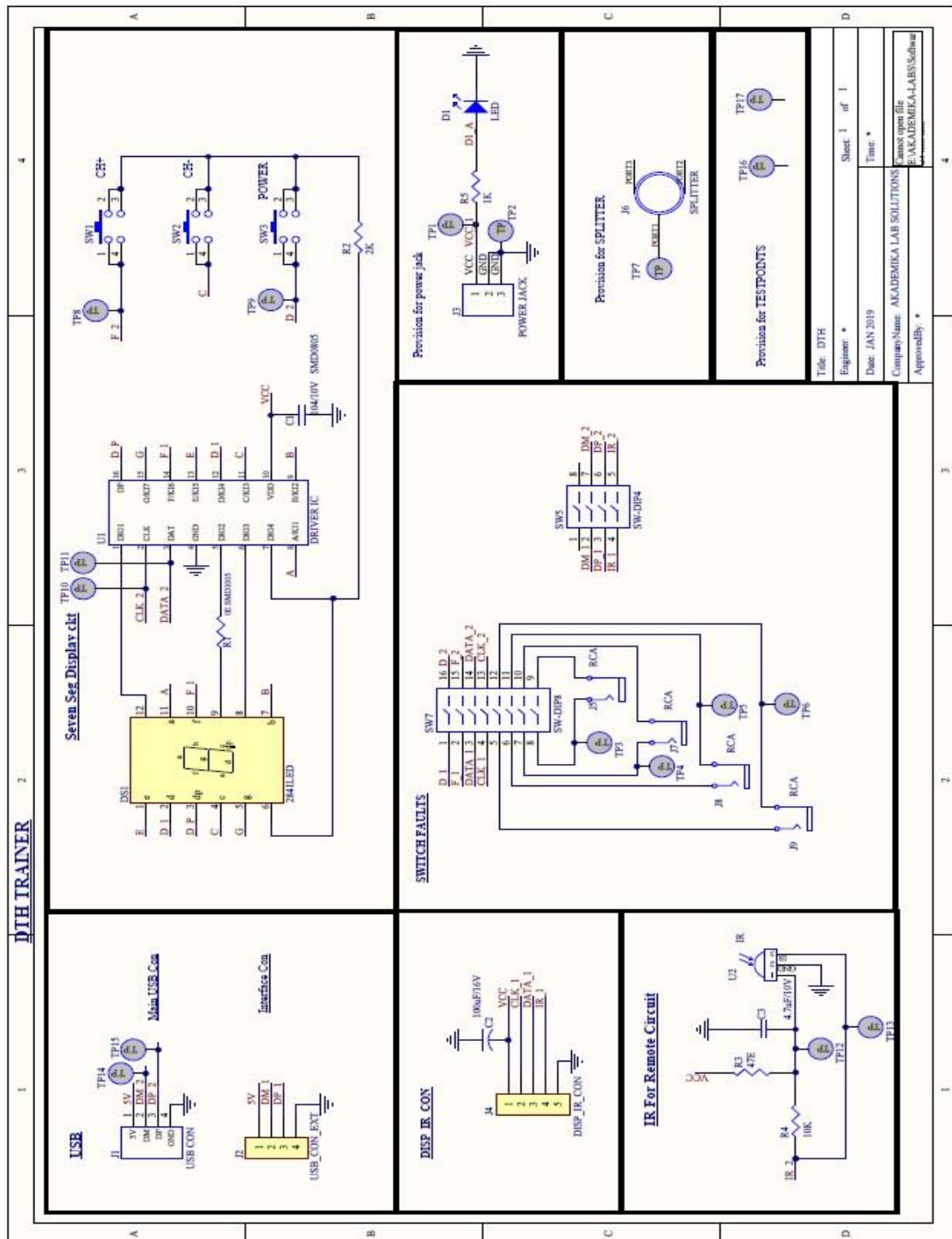
Deliverables

- | | |
|---|-----------|
| 1) Remote with pencil cell | : 1 No |
| 2) Dish Antenna with mounting Accessories | : 1 No |
| 3) Power Adapter | : 1 No |
| 4) Dish Connecting Wire | : 12 mtr. |
| 5) Universal LNB | : 1 No |
| 6) RCA male to RCA male cable | : 1 No |
| 7) Product user Manual | : 1 No |

Functional Block Diagram



Circuit Diagram



**EXPERIMENT
NO. 01**

EXPERIMENT NO: 01

NAME

Introduction and basics of DTH system.

OBJECTIVE

To learn Introduction and basics of DTH system.

THEORY

DTH stands for Direct-To-Home television. DTH is defined as the reception of satellite programmes with a personal dish in an individual home.

DTH does away with the need for the local cable operator and puts the broadcaster directly in touch with the consumer. Only cable operators can receive satellite programmes and they then distribute them to individual homes.

Direct-to-Home (DTH) television is a method of receiving satellite television by means of signals transmitted from direct-broadcast satellites. The Government of India permitted the reception and distribution of satellite television signals in November 2000. The first DTH service in the country was launched by Dish TV on 2 October 2003. DD Free Dish, the first free DTH service in India, was launched by public broadcaster Prasar Bharati in December 2004.

India is the largest DTH market in the world by number of subscribers. As on 31 December 2017, there were 67.56 million active pay DTH subscribers in the country. These figures do not include subscribers of free DTH services. The market is serviced by 4 pay DTH providers and one free DTH provider.

DTH services were first proposed in India in 1996. The proposal was not approved to due to concerns over national security and negative cultural influence. In 1997, the Government of India banned DTH services when Rupert Murdoch-owned Indian Sky Broadcasting (ISkyB) was about to launch its DTH services in the country. After deliberations among groups of ministers, DTH services were permitted by the NDA government in November 2000. The ministers made four key recommendations to govern DTH services: no single entity, either private or state-owned, should be permitted a monopoly in DTH services; the vertical integration of DTH and cable television services should be monitored in order to prevent the formation of a monopoly in television distribution; the vertical integration of DTH operators and television channels should be avoided in order to ensure fair competition among TV channels.

The new policy announced in November 2000 required all DTH operators to set up earth stations in India within 12 months of receiving a license. DTH licenses were priced at \$2.14 million with a validity of 10 years. The FDI limit in the DTH sector was

capped at 49%, and the company operating the services was required to be headed by an Indian citizen. The first DTH service was launched in India on 2 October 2003 by Dish TV owned by Zee. The company decided not to compete against entrenched cable operators in metros and urban areas, and instead focused on providing services to rural areas and regions not serviced by cable television. Dish TV acquired 350,000 subscribers within 2 years of the launch. Public broadcaster Prasar Bharati launched DD Direct Plus (now DD Free Dish) in December 2004. The service is free and offers only free-to-air channels.[5] Tata Sky was incorporated in 2004 as joint venture between the Tata Group and Star India's Parent Company. Tata Sky launched DTH services in August 2006. Unlike Dish TV, Tata Sky focused on metros and large cities hoping to attract customers away from cable by offering better picture and audio quality and wider selection of channels.[4] Following legal proceedings between STAR and Zee, in 2007, the two companies called a truce and began offering their channels on each other's services. This decision and Dish TV's acquisition of more transponders enabled them to offer 150 channels on their service, more than any other DTH service in India at the time.[4]

Sun Direct and Airtel digital TV launched services in 2007 and 2008 respectively. Reliance Big TV (now Independent TV) was launched in August 2008.[6] The service acquired 1 million subscribers within 90 days of launch, the fastest ramp-up ever achieved by any DTH operator in the world.[7] Videocon d2h launched its services in June 2009.

The total number of DTH subscribers in India rose from 1.5 million in 2005 to 23 million in 2010.[8] Sun Direct became the first DTH provider to offer high-definition (HD) channels in early 2010.[9] Tata Sky began offering HD channels later that year.[10] Other DTH providers subsequently began carrying HD channels.

List of DTH Providers

S. No.	Service Provider	Launch date	Subscribers May 2018	Ownership
1	- Dish TV - d2h ^[39] - Zing Digital	October 2003 June 2009 January 2015	15.6 million 12.5 million	Essel Group
2	Tata Sky	August 2006	14.3 million	Tata Sons (60%) 21st Century Fox (30%) Temasek Holdings (10%)
3	Airtel digital TV	October 2008	12.5 million	Bharti Airtel Limited (80%) Warburg Pincus (20%) ^[38]
4	Sun Direct	2007	5.2 million	Sun Network (80%) Astro Group (20%)
5	DD Free Dish	December 2004	35 million ^[40]	Prasar Bharati
6	Independent TV	March 2018	15.50 million	Pantel Technologies Pvt Ltd. (50%) Veecon Media & Television Limited (50%)

Satellites

The Department of Space (DoS) requires all DTH operators in India to only use satellites commissioned by the Indian Space Research Organization (ISRO). DTH operators may use capacity leased by ISRO from foreign satellites only if sufficient capacity is not available on ISRO satellites. The following 8 satellites are currently in use by Indian DTH service providers

Satellite	Owner	Mission life	User(s)
INSAT-4A	ISRO	12 years	Freeview
GSAT-10	ISRO	15 years	Freeview
ST-2	Singtel	15 years	Freeview
MEASAT-3	MEASAT Satellite Systems	15 years	Freeview
GSAT-15	ISRO	12 years	Freeview
NSS-6	SES S.A.	15 years	Freeview
AsiaSat 5	AsiaSat	17 years	Freeview
GSAT-17	ISRO	15 years	Freeview

Compression standards

All DTH services in India currently use the MPEG-4 standard of signal compression. MPEG-2 is still used by DishTV, TATA Sky, DD Free Dish. Upgradation is going on to shift from MPEG2 to MPEG4, but to shift completely from MPEG2 to MPEG4, the subscriber have to change his/her STB 1st because MPEG2 STB can not decode MPEG4 video signal. MPEG-2 permitted each transponder to carry approximately 20 SD channels (fewer, in case of HD channels), while MPEG-4 enables each transponder to carry approximately 50 SD channels (again, fewer in case of HD channels). The bandwidth required to carry a single HD channel is approximately equivalent to the bandwidth required to carry 4 SD channels.

**EXPERIMENT
NO. 02**

EXPERIMENT NO: 02

NAME

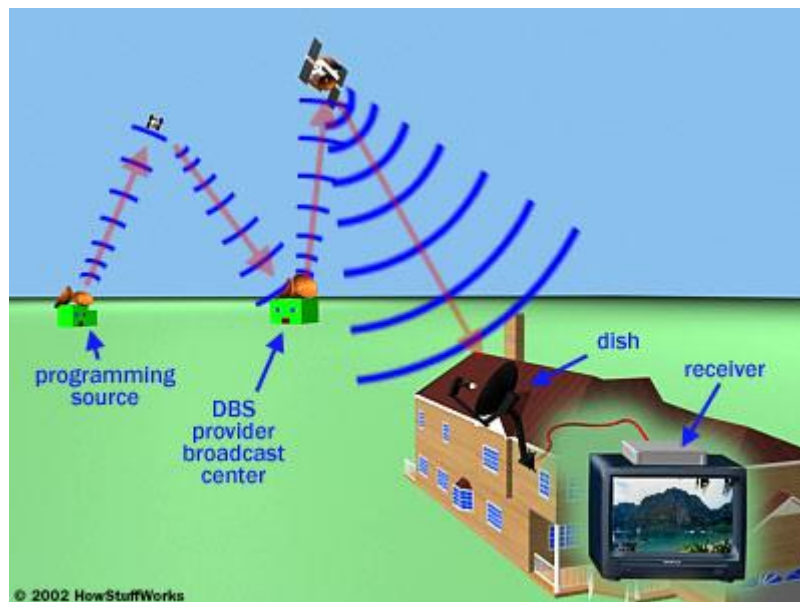
Understanding various components of DTH Receiver system.

OBJECTIVE

To Understanding various components of DTH Receiver system.

THEORY

There are five major components involved in a direct to home (DTH) or direct broadcasting (DBS) satellite system: the programming source, the broadcast center, the satellite, the satellite dish and the receiver.



Programming sources are simply the channels that provide programming for broadcast. The provider doesn't create original programming itself; it pays other companies (HBO, for example, or ESPN) for the right to broadcast their content via satellite. In this way, the provider is kind of like a broker between you and the actual programming sources. (Cable TV companies work on the same principle.)

The broadcast center is the central hub of the system. At the broadcast center, the TV provider receives signals from various programming sources and beams a broadcast signal to satellites in geosynchronous orbit.

The satellites receive the signals from the broadcast station and rebroadcast them to Earth.

The viewer's dish picks up the signal from the satellite (or multiple satellites in the same part of the sky) and passes it on to the receiver in the viewer's house.

The receiver processes the signal and passes it on to a standard TV.

Satellite TV Programming

Satellite TV providers get programming from two major sources: national turnaround channels (such as HBO, ESPN and CNN) and various local channels (the ABC, CBS, Fox, NBC and PBS affiliates in a particular area). Most of the turnaround channels also provide programming for cable TV, and the local channels typically broadcast their programming over the airwaves.

Turnaround channels usually have a distribution center that beams their programming to a geosynchronous satellite. The broadcast center uses large satellite dishes to pick up these analog and digital signals from several sources.

Most local stations don't transmit their programming to satellites, so the provider has to get it another way. If the provider includes local programming in a particular area, it will have a small local facility consisting of a few racks of communications equipment. The equipment receives local signals directly from the broadcaster through fiber-optic cable or an antenna and then transmits them to the central broadcast center.

The broadcast center converts all of this programming into a high-quality, uncompressed digital stream. At this point, the stream contains a vast quantity of data -- about 270 megabits per second (Mbps) for each channel. In order to transmit the signal from there, the broadcast center has to compress it. Otherwise, it would be too big for the satellite to handle. In the next section, we'll find out how the signal is compressed.

Satellite TV Signal

Satellite signals have a pretty long path to follow before they appear on your TV screen in the form of your favorite TV show. Because satellite signals contain such high-quality digital data, it would be impossible to transmit them without compression. Compression simply means that unnecessary or repetitive information is removed from the signal before it is transmitted. The signal is reconstructed after transmission.

Standards of Compression

Satellite TV uses a special type of video file compression standardized by the Moving Picture Experts Group (MPEG). With MPEG compression, the provider is able to transmit significantly more channels. There are currently five of these MPEG standards, each serving a different purpose. DirecTV and DISH Network, the two major satellite TV providers in the United States, once used MPEG-2, which is still used to store movies on DVDs and for digital cable television (DTV). With MPEG-2, the TV provider can reduce the 270-Mbps stream to about 5 or 10 Mbps (depending on the type of programming).

Now, DirecTV and DISH Network use MPEG-4 compression. Because MPEG-4 was originally designed for streaming video in small-screen media like computers, it can

encode more efficiently and provide a greater bandwidth than MPEG-2. MPEG-2 remains the official standard for digital TV compression, but it is better equipped to analyze static images, like those you see on a talk show or newscast, than moving, dynamic images. MPEG-4 can produce a better picture of dynamic images through use of spatial (space) and temporal (time) compression. This is why satellite TV using MPEG-4 compression provides high definition of quickly-moving objects that constantly change place and direction on the screen, like in a basketball game.

MPEG STANDARDS

All MPEG standards exist to promote system interoperability among your computer, television and handheld video and audio devices. They are:

MPEG-1: the original standard for encoding and decoding streaming video and audio files.

MPEG-2: the standard for digital television, this compresses files for transmission of high-quality video.

MPEG-4: the standard for compressing high-definition video into smaller-scale files that stream to computers, cell phones and PDAs (personal digital assistants).

MPEG-21: also referred to as the Multimedia Framework. The standard that interprets what digital content to provide to which individual user so that media plays flawlessly under any language, machine or user conditions.

**EXPERIMENT
NO. 03**

EXPERIMENT NO: 03

NAME

Installation of Dish Antenna

OBJECTIVE

To study the installation and set up of satellite TV program receiver system.

THEORY

How to Install and Set Up Free to Air Satellite TV Program Receiver System

Select a satellite network. In order to check your satellite's compatibility with your current location, you'll have to know the name of the satellite itself.

Make sure that you can receive your satellite's signal. Before attempting to set up an FTA system, you should know if it's possible for you to receive your selected satellite's signal. You can test this by going to <http://www.dishpointer.com/> and doing the following:

- Type your city and state (e.g., "Palo Alto, California") into the "Your location" text box on the left side of the page.
- Select your satellite's name from the drop-down box on the right side of the page.
- Click Search!
- Make sure that there is a green line indicating line-of-sight for your satellite. If this line is red, the satellite won't work properly in your area.

Note the network's bearings. In the pop-up box on the map, look at both the "Elevation" number and the "Azimuth (true)" number. You'll use these numbers (in degrees) to adjust your dish later.

Ensure that you have the proper hardware. You'll need the following equipment before you can actually install the satellite dish:[1]

- Satellite dish — Used to receive the satellite's signal. You'll need an 8-foot dish for C-band channels or a 35-inch dish for KU-band channels.
- Satellite receiver — Used to receive the satellite dish's input and translate it into channels for your TV.
- Satellite tuner — Used to fine-tune the satellite dish's position.
- HDTV — Necessary for most FTA systems, as the receiver will generally require an HDMI input on the TV itself.
- Coaxial cable — Usually packaged with the satellite dish, but you may need to buy a longer or shorter cable depending on your satellite dish's location.

Determine your satellite dish's optimal position. You'll want to aim your satellite dish at the satellite itself, so find a high location (e.g., a rooftop or a balcony) from which the dish can face the satellite without losing track of the satellite's bearings.

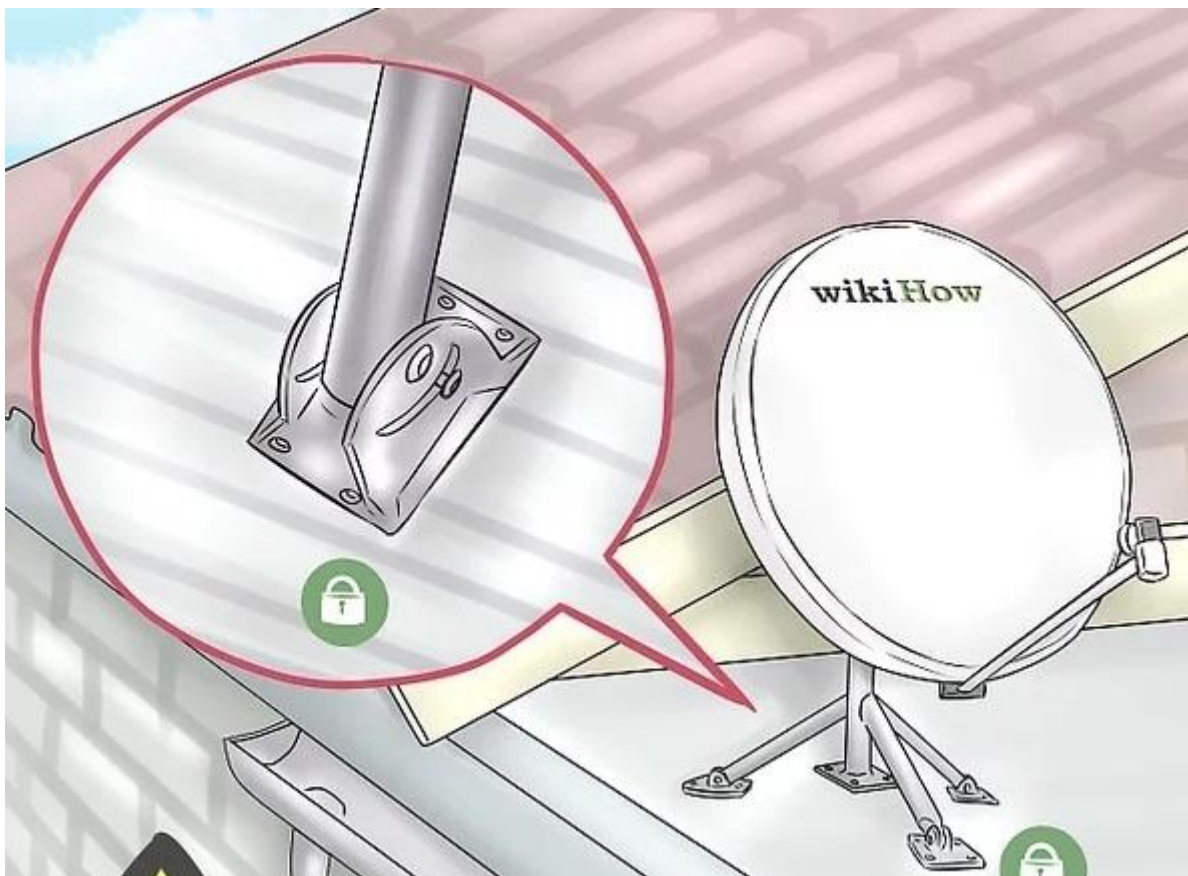
- You'll also need to ensure that your satellite dish isn't obstructed by trees, buildings, or other obstacles.

Figure out the best route from your dish to your TV. Since you'll need to run a coaxial cable from the dish to the receiver inside your house, you'll want to find a path that minimizes the cable's exposure to the elements while keeping the cable as short as possible.

- Many dish users route the cable along the side of the house and through a wall where necessary, but your cable setup may vary.
- If necessary, buy a new coaxial cable that will reach your dish from the receiver before proceeding.

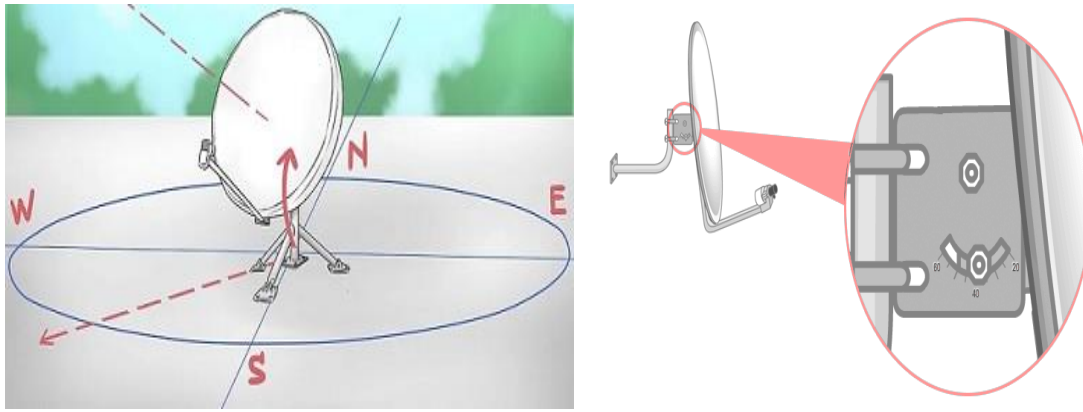
Part2

Installing the Receiver Dish



Secure the dish in your selected position. Place the dish's mast and dish itself on a stable surface, then lock down its position using the included bolts or fasteners.

- It's incredibly important to make sure that the dish is secured as tightly as possible so that it doesn't become loose during a windstorm.
- If the dish is mounted on a wood roof, you can caulk around the base to make it waterproof.



Point the dish toward your satellite. Using the "Elevation" and "Azimuth" numbers as guides, angle your satellite dish toward the satellite that you want to use. This will ensure that the dish is roughly positioned toward the satellite.

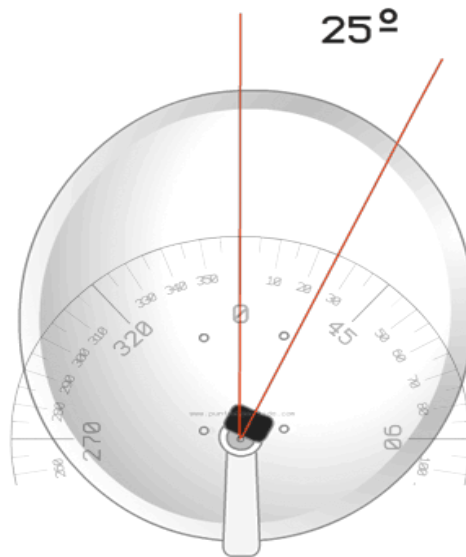
Connect the satellite tuner. Using the 6 foot (1.8 m)-long coaxial cable on your satellite, plug in the satellite tuner.

Use the satellite tuner to fine-tune your dish's horizontal axis. Turn on the satellite finder, input the satellite name or select it from a list, and enter the satellite's frequency. You should hear a continuous beeping sound which will allow you to corroborate your dish's position:

- Rotate the dish left or right.
- Listen for quicker beeping to ensure that you're rotating the dish in the correct direction.
- Rotate the dish the other way if the spaces between beeps get longer.

Secure your dish's horizontal axis. Tighten the rotation control screw completely to fix this angle.

Adjust the vertical axis. You'll do this in the same way as you adjusted the horizontal axis; once the beeping is as quick as you can manage to achieve, you can tighten the vertical axis screw.



Connect your satellite dish to your receiver. You'll use the long coaxial cable for this step. The coaxial cable should plug into the back of the satellite dish's receiver.

- You can use a staple gun to staple the coaxial cable to your house's siding in order to prevent it from hanging freely.
- Depending on your house's layout, you may need to drill a hole in the wall in order to thread your coaxial cable through to the receiver. If so, make sure not to drill through any pipes or wires.

Part3

Setting up the Receiver

Plug the receiver into both a power source and your TV. Once you've attached the coaxial cable to the receiver, you can use the receiver's HDMI cable to attach to one of your TV's HDMI ports.

- You'll also need to use the receiver's power cable to connect to an electrical outlet.

Turn on the receiver if necessary. Your receiver should turn on when plugged in, but there may be an On/Off switch on the side or the back of the receiver. If so, flip the switch to the "On" position before proceeding.

Switch to the receiver's channel. Turn on your TV, then switch to the HDMI input into which you plugged the receiver.

- For example, if you plugged your receiver into the "HDMI 1" slot, you'll switch the input to the "HDMI 1" channel using your TV's Input or Video menu.

Allow the receiver to perform its installation if necessary. Some receivers will go through an automatic setup process the first time they're turned on; if so, allow your receiver to complete the setup before continuing.

- If prompted to perform any actions during the setup, follow the on-screen instructions.

Open the receiver's menu. On your receiver's remote, find and press the Menu button. You should see a pop-up menu appear on-screen.

Find your dish's antenna setup menu. You'll usually have to use your remote's arrow buttons to find the "Install" or "Dish" option, but consult your receiver's manual if you can't find the setup section of the menu. Select a satellite. In the "Satellite" section of the menu, use the arrows to scroll left or right until you find your satellite's name.

Select an LNB frequency. In the "LNB" section of the menu, use the arrows to select 10750 as the LNB number. This is the most commonly used LNB frequency for satellite networks.[2]

- If you're using a C-band network, you'll select 5150 here instead.

Scan for channels. Find the "Scan" or "Single Satellite Scan" section of the menu, set the "FTA Only" section to Yes if possible, and start the scan by selecting Yes, OK, or Start. Your dish will begin searching for available satellite TV channels; once it finishes, you'll be able to watch TV like usual on your dish's channel.

**EXPERIMENT
NO. 04**

EXPERIMENT NO: 04

NAME

Satellite Finder.

OBJECTIVE

Study of Satellite finder meter .

THEORY

Satellite finder meter

A satellite finder meter is a specialized device designed to help you position your satellite dish to receive the strongest signal for best television reception. They are simple to operate and are designed to be used while you're at the satellite dish so you don't have to check your signal strength at your television set. There are both analog and digital models available, both with and without internal power supplies.

Step 1

Connect one end of the coaxial cable to the barrel connector on the satellite dish's low noise block downconverter (LNB) and the other end to the satellite finder meter. The LNB is the arm that faces the reflector dish.

Step 2

Loosen the nuts that allow the dish to be adjusted both along the horizontal (azimuth) and vertical (elevation) planes.

Step 3

Move the dish from side to side along the horizontal plane until you determine the position at which the satellite finder meter displays maximum signal strength.

If you're using an analog meter, you will find this position in increments. First, adjust the gain knob so the meter's needle reads half strength. As you move the dish closer to its optimal position, the meter may exceed maximum. If this happens, readjust the gain lower and continue to move the dish until you determine the position where maximum signal strength is received.

Step 4

Raise and lower the dish along the vertical plane until you determine the elevation at which the satellite finder meter displays maximum signal strength. If using an analog meter, you may have to adjust the gain as in the previous step.

Step 5

Tighten the nuts to lock your satellite dish into place.

Step 6

Disconnect the coaxial cable from the LNB on the satellite dish.



**EXPERIMENT
NO. 05**

EXPERIMENT NO: 05

NAME

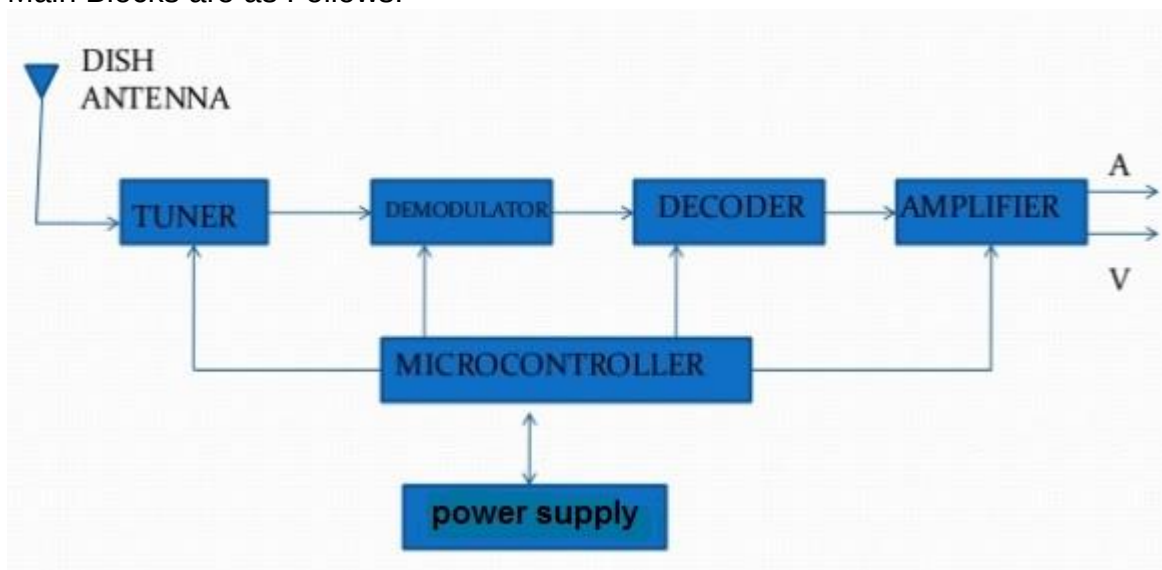
Block diagram of DTH

OBJECTIVE

Study of various blocks of DTH system.

THEORY

Main Blocks are as Follows:



1) LNB:

A low-noise block downconverter (LNB) is the receiving device mounted on satellite dishes used for satellite TV reception, which collects the radio waves from the dish and converts them to a signal which is sent through a cable to the receiver inside the building. Also called a low-noise block, low-noise converter (LNC), or even low-noise downconverter (LND), the device is sometimes inaccurately called a low-noise amplifier (LNA).

The LNB is a combination of low-noise amplifier, frequency mixer, local oscillator and intermediate frequency (IF) amplifier. It serves as the RF front end of the satellite receiver, receiving the microwave signal from the satellite collected by the dish, amplifying it, and down converting the block of frequencies to a lower block of intermediate frequencies (IF). This down conversion allows the signal to be carried to the indoor satellite TV receiver using relatively cheap coaxial cable; if the signal remained at its original microwave frequency it would require an expensive and impractical waveguide line.

The LNB is usually a small box suspended on one or more short booms, or feed arms, in front of the dish reflector, at its focus (although some dish designs have the LNB on or behind the reflector). The microwave signal from the dish is picked up by

a feed horn on the LNB and is fed to a section of waveguide. One or more metal pins, or probes, protrude into the waveguide at right angles to the axis and act as antennas, feeding the signal to a printed circuit board inside the LNB's shielded box for processing. The lower frequency IF output signal emerges from a socket on the box to which the coaxial cable connects.

LNB disassembled. The waveguide from the feed horn, carrying the microwave radio signal collected by the dish passes through the hole in the center. The pins visible at the top and left side of the hole project into the waveguide and act as antennas to receive the signal, converting it to radio frequency alternating currents which are processed by the circuit board. The lower frequency output signal is taken from the coaxial cable jacks, bottom.

Cross-section across a low-noise block downconverter.

Viewing of the pin and the horn antenna in a low-noise block downconverter.

The LNB gets its power from the receiver or set-top box, using the same coaxial cable that carries signals from the LNB to the receiver. This phantom power travels to the LNB; opposite to the signals from the LNB.

A corresponding component, called a block up converter (BUC), is used at the satellite earth station (uplink) dish to convert the band of television channels to the microwave uplink frequency.

2) Tuner:-

The ATSC tuner module is available for accessing QAM-, OFDM, and QPSK-based networks. In addition to receiving inputs from digital networks, most tuners are also capable of tuning analog broadcasts. Tuners can be divided into three broad categories:

'''Broadcast In-band (IB) Tuner.''' Once the signal arrives from the physical transmission media, the IB tuner will isolate a physical from a multiplex of channels and convert to baseband. The term "baseband" is used to describe a single channel or digital signal, extracted from broadband signal which is a stream of multiple channels.

'''Out Of Band (OOB) Tuner.''' This type of tuner facilitates the transfer of data between the head-end systems and the set-box. They are widely used in cable set-top boxes for providing subscribers with medley of interactive services. Implementations of the OOB tuner tend to operate within 100 to 350 MHz frequency band.

'''Return Path Tuner.''' This tuner allows a subscriber to activate the return of path and send data back to the interactive services provider. Implementations of this tuner tend to operate within the 5 to 60 MHz frequency band.

3) Demodulator:

The function of the demodulator is to convert the analog signal to a digital bit stream. Then, it checks for errors and forward the bit stream to the demultiplexer.

4) Decoder:

A DTV receiver will normally contain three separate decoders for converting the digital bit-stream back into a format that can be heard and viewed by the subscriber.

A video decoder''' It transforms video packets into a sequence of pictures, which are displayed by the TV monitor. An audio decoder''' It decompresses the audio bit-stream. Different audio modes are usually supported by the DTV receiver: mono and dual channel, stereo, and joint stereo. A data decoder.

5) Amplifier:

The purpose of an amplifier is to receive a small electrical signal and enlarge or amplify it. In the case of a pre-amplifier, the signal must be amplified enough to be accepted by a power amplifier. In the case of a power amplifier, the signal must be enlarged much more, enough to power a speaker.

6) Central processing unit (controller):

The CPU, like in a desktop PC, is the brain of the DTV receiver. Functions provided by a CPU include:

- initializes the various hardware components;
- processes a range of Internet and interactive TV applications;
- monitors and manages hardware interrupts;
- fetches data and instructions from memory;
- runs various programs.

**EXPERIMENT
NO. 06**

EXPERIMENT NO: 06

NAME

Working Principle of DTH.

OBJECTIVE

Study of working principle of DTH system.

THEORY

Direct to home (DTH) television is a wireless system for delivering television programs directly to the viewer's house. In DTH television, the broadcast signals are transmitted from satellites orbiting the Earth to the viewer's house. Each satellite is located approximately 35,700 km above the Earth in geosynchronous orbit. These satellites receive the signals from the broadcast stations located on Earth and rebroadcast them to the Earth.

Early satellite TV viewers were explorers of sorts. They used their expensive dishes to discover unique programming that wasn't necessarily intended for mass audiences. The dish and receiving equipment gave viewers the tools to pick up foreign stations, live feeds between different broadcast stations, NASA activities and a lot of other stuff transmitted using satellites. Some satellite owners still seek out this sort of programming on their own, but today, most Direct to home TV customers get their programming through a direct broadcast satellite (DBS) provider, such as DirecTV or the Dish Network. The provider selects programs and broadcasts them to subscribers as a set package. Basically, the provider's goal is to bring dozens or even hundreds of channels to your television in a form that approximates the competition, cable TV. Unlike earlier programming, the provider's broadcast is completely digital, which means it has much better picture and sound quality. Early satellite television was broadcast in C-band radio -- radio in the 3.4-gigahertz (GHz) to 7-GHz frequency range. Digital broadcast satellite transmits programming in the Ku frequency range (12 GHz to 14 GHz).

**EXPERIMENT
NO. 07**

EXPERIMENT NO: 07**NAME**

Switch faults in DTH receiver.

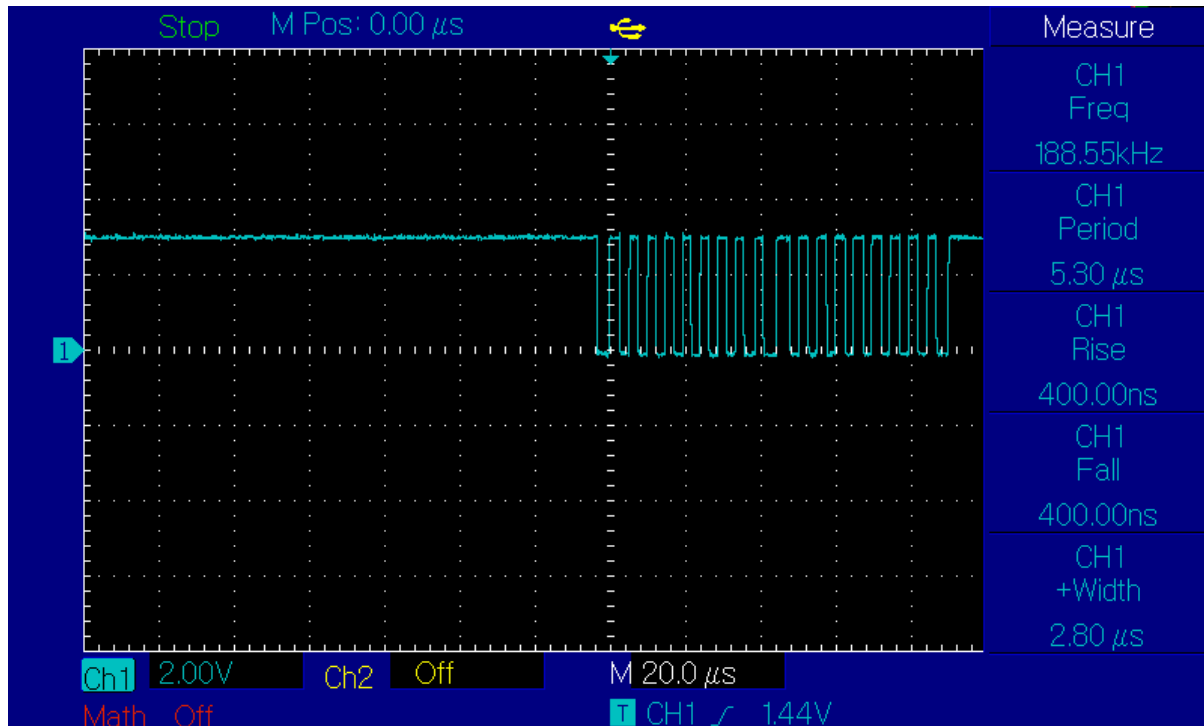
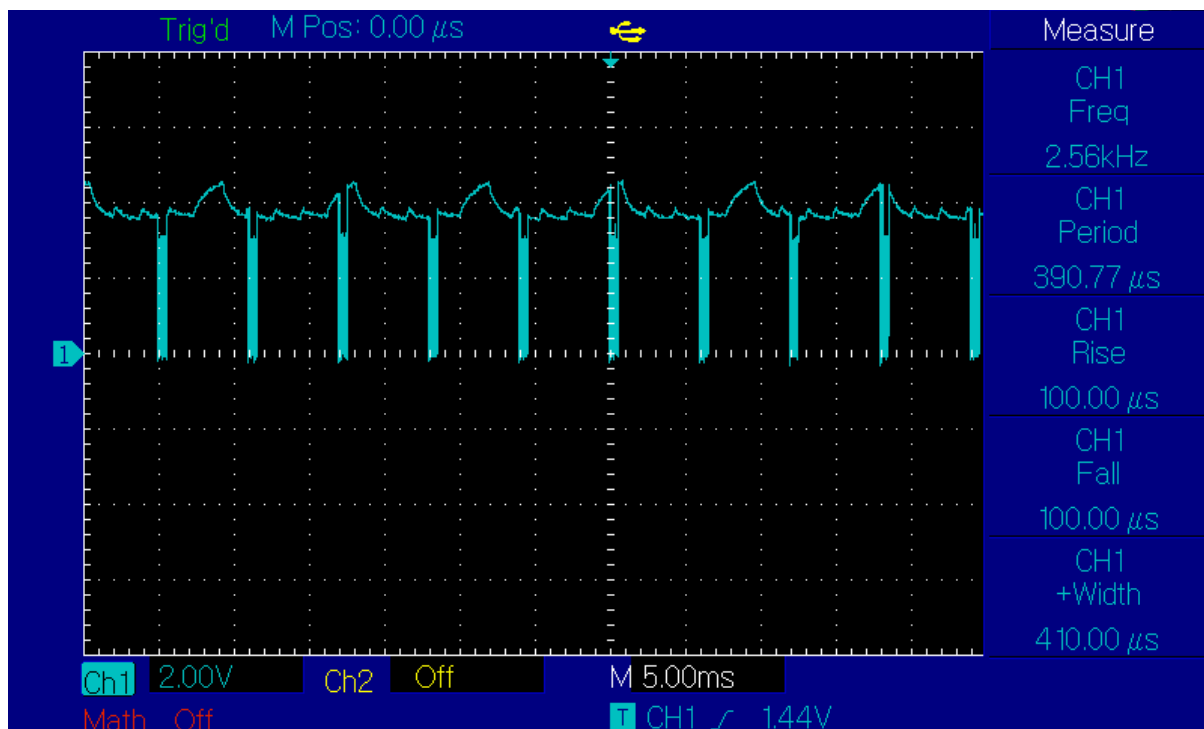
OBJECTIVE

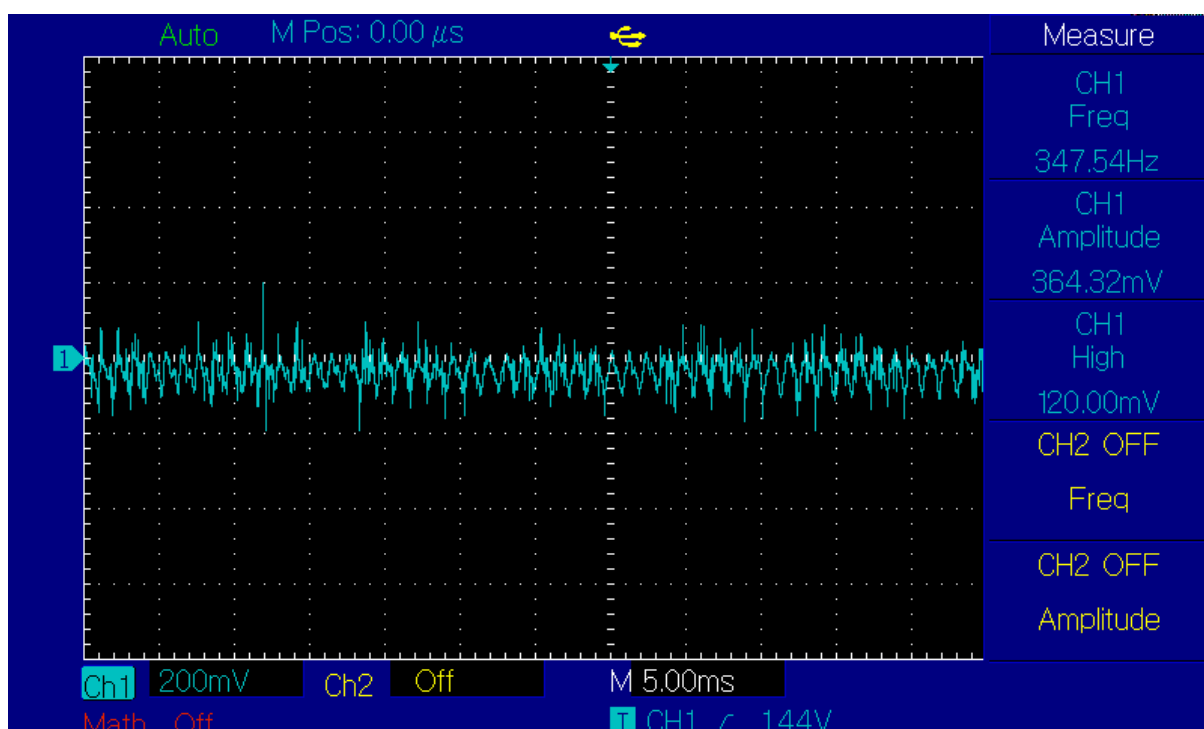
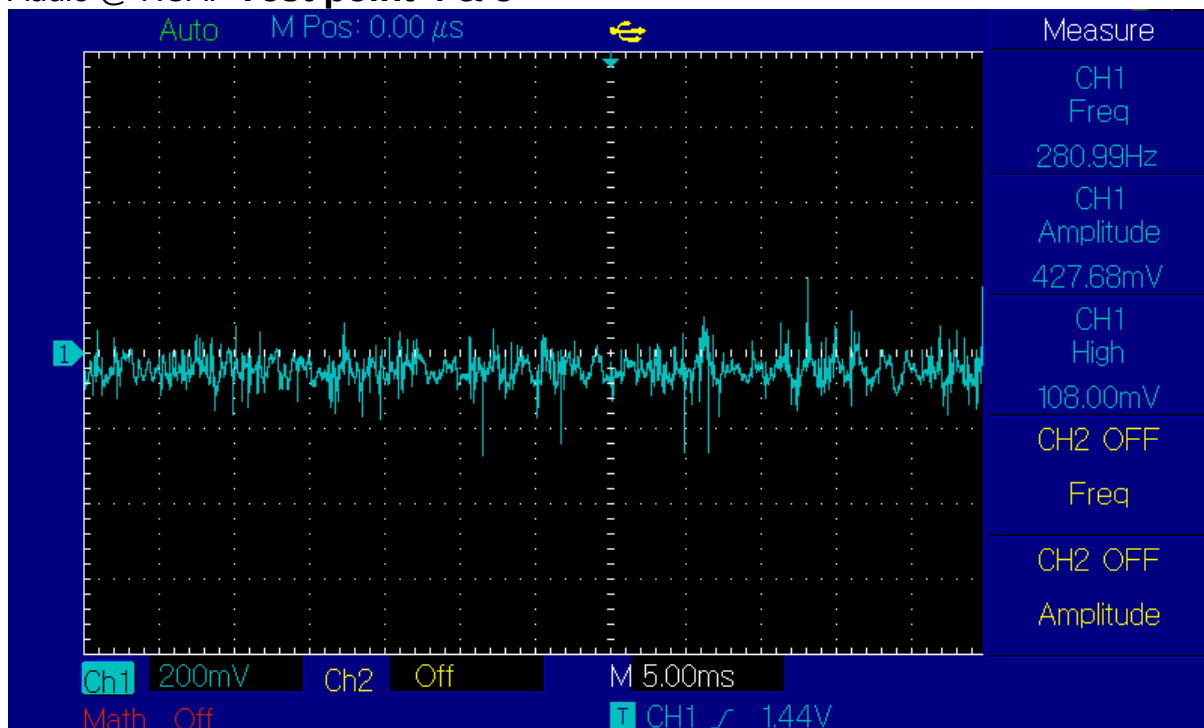
Study of various switch faults and observe the waveforms of DTH.

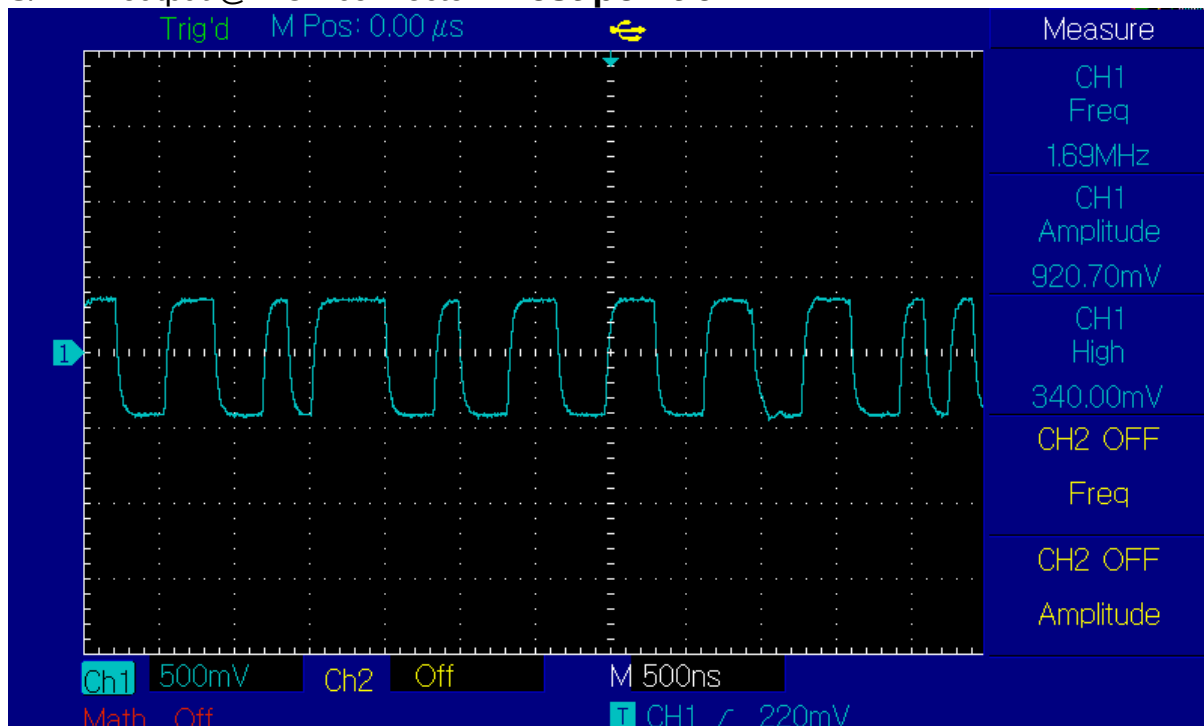
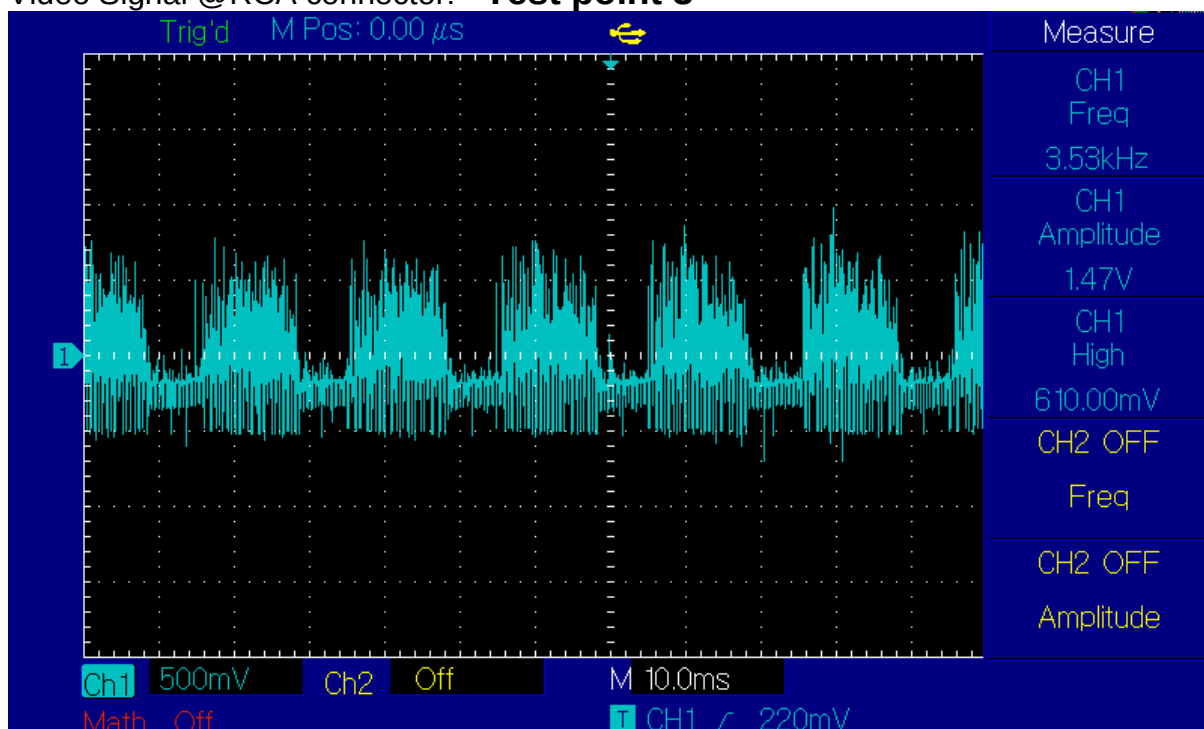
Switch Faults

SR NO	SWITCH NO	FUNCTION
1	SW 7, 1	Power Switch from keypad not Working.
2	SW 7, 2	Channel CH+ Switch from keypad not Working.
3	SW 7, 3	Display dose not change the channel no due to Data not received from processor.
4	SW 7, 4	Display dose not change the channel no due to clock not received to display driver IC.
5	SW 7, 5	Video is disabled.
6	SW 7, 6	Audio is disabled (left).
7	SW 7, 7	Audio is disabled (Right).
8	SW 7, 8	S/PDIF (Sony/Philips Digital Interface) is disabled.
9	SW 5, 2	Data M of USB is disabled.
10	SW 5, 3	Data P of USB is disabled.
11	SW 5, 4	IR sensor not working (remote is not working)

Test waveforms

Clock Frequency To Display:- **Test point 10**IR Receiver:- **Test point 13**

Audio @ RCA:- **Test point 4 & 5**

S/PDIF output @ RCA connector:- **Test point 6**Video Signal @RCA connector:- **Test point 3**

Output of LNB To STB:- **Test point 7**
(For this you need Spectrum analyzer)

